STAMATIS CHOUDALAKIS

Athens, Greece | +30 6985793600 | st.xoud@gmail.com | https://stchoud.github.io/ | https://github.com/Stchoud

SUMMARY

Mathematician with a strong foundation in data management principles seeking a Data Engineer role. With proven 5+ years of experience in designing, developing, and optimizing robust ETL pipelines and data processing workflows using Python and R. Skilled in identifying and resolving data flow bottlenecks and data quality issues, and in building clean, rigorously tested code. Eager to leverage my strong analytical background and problem-solving skills to contribute to efficient and scalable data infrastructure. Military obligations are fulfilled.

EXPERIENCE

Freelancing

Athens, Greece — Sep 2020 – Present

Provided comprehensive courses in diverse mathematical fields to over 80 undergraduate and postgraduate students. I also guided students through their assignments in Python, R, and SQL, covering aspects of development and data science projects, including data curation, web scraping, statistical analysis, mathematical modelling, and effectively communicated complex technical subjects in an accessible manner.

Elementary School Teacher

Zygouraki-Leontioy Centre of Creative Activities for Children | Athens, Greece — Jan 2024 – Jun 2024

Tutored and supported 8+ elementary students per hour in completing math assignments.

Ballroom Dance Instructor

The Dance Club, The Mall of Dance | Athens, Greece — Sep 2016 – Jul 2022

Taught, choreographed, and competed in Latin, European, and Modern ballroom dances with students aged 15-65. Mentored 5 students who are now professional dancers.

EDUCATION

PhD in Bioinformatics

Mathematics Research Center, Academy of Athens| School of Medicine, NKUA — Mar 2024 – Present

Thesis: ``Cancer driver gene identification using machine learning".

Identified 42 previously unreported mutational patterns in cancer initiation and progression. Built state-of-the-art automated machine learning and data curation pipelines that handled diverse and complex biological data using Python and R in a custom Docker container. Streamlined and optimized complex data processing and computational workflows and solved critical bottlenecks. Created rigorously tested and clean code for reproducible results.

MSc in Applied Mathematics (Grade: 8.11/10)

Department of Mathematics, NKUA — Sep 2020 – Jun 2022

Syllabus included advanced topics in statistical simulation, statistics of machine learning, numerical methods for ODEs, and numerical linear algebra. Dissertation: ``Network Graphs of Cancer Mutations" in https://pergamos.lib.uoa.gr/uoa/dl/object/3237319.

BSc in Mathematics (Grade: 6.29/10)

Department of Mathematics, NKUA — Sep 2015 – Sep 2020

LECTURES

Lectures on Machine Learning
School of Medicine, NKUA — Mar 2024 – Present

PROFESSIONAL CERTIFICATES

IBM Data Engineering
IBM| Coursera — Sep 2025

Professional Certificate in Data Engineering, with coursework in Python for Data Science, Relational Databases & SQL, ETL & Data Pipelines, Data Warehousing, Big Data (PySpark, Spark SQL, Hadoop), Machine Learning, BI Dashboards (IBM Cognos Analytics, Google Looker), NoSQL (MongoDB, Cassandra), and orchestration (Apache Airflow).

Snowflake Data Engineering
Snowflake| Coursera — In Progress

Currently pursuing a Professional Certificate in Data Engineering with Snowflake gaining hands-on expertise in Snowflake for data development, science, and engineering.

TECHNICAL SKILLS

Programming Languages

- Python (ETL, data visualization, machine learning, web scraping)
- R (ETL)
- Bash (task automation, inter-language integration)
- SQL (MySQL, PostrgreSQL - actively developing proficiency)

Code is freely accessible upon request:

- https://github.com/Stchoud/patterns_intra_clustering
- https://github.com/Stchoud/multiomic_intra_clustering.

Tools

VS Code, Git, Docker, Jupyter Notebook, LaTeX, SSH.

Cloud & Data Platforms

Building foundational knowledge in Snowflake, Databricks, and AWS through coursework.

PUBLICATIONS

Choudalakis S., Kastis G. A., & Dikaios N. (2025). **Intra-clustering analysis reveals tissue-specific mutational patterns**. Computer Methods and Programs in Biomedicine, 263, 108681. (https://doi.org/10.1016/j.cmpb.2025.108681)

Choudalakis S., Mitrouli M., Polychronou A., & Roupa P. (2021). **Solving high-dimensional problems in statistical modelling: a comparative study**. Mathematics, 9(15), 1806. (https://doi.org/10.3390/math9151806)

FOREIGN LANGUAGES

English
Level C2, ECPE, University of Michigan — Jun 2013

Italian
Level B1, National Foreign Language Exam System — Jun 2012